

Review retrospective:
*the multivariate change-of-variables bridge — a clean 1547-line
pass, redundant opens, and an affirmation that caught my own
corner case*

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Abstract

A soundness review of `Laplace/Multi/RescaledIntegrals.lean` (1547 lines, ~60 lemmas): the multivariate change-of-variables bridge that relates the Gibbs partition / numerator / expectation / covariance to their *rescaled* forms under the dilation $x \mapsto R \cdot x$, together with the coercivity, integrability, and Gaussian-domination infrastructure the multivariate covariance asymptotics consume. Result: a clean fidelity pass; the landables were a dead **open**, six *redundant* per-section **opens**, three dead equation-names, and two docstring over-claim fixes — one of which the affirmation pass corrected a corner case in.

1 Setting

The first of the large `Multi/` analytic files (after the smaller `Multi/` pieces were reviewed earlier in the marathon), and the substrate beneath the `Multi/Covariance{,Sharp,Explicit}` trio. It carries the Stage-3 machinery: rescale to a unit-variance Gaussian, dominate the perturbation uniformly in t , and integrate.

2 What was reviewed

The dilation bridge — `integral_comp_smul_pi`, `rescaledNumerator_eq_smul`, `rescaledPartition_eq_smul`, and the headline cancellations `gibbsExpectation_eq_rescaledExpectation / gibbsCov_eq_rescaledCov` — plus the coercivity bounds (`quadForm_lower_bound`, `rescaled_weight_le_coercive`), the integrability lemmas (`rescaled_weight`, `polynomial×Gaussian` decay, `coordinate/norm` moments), and the elementary norm/sum-of-squares helpers. Priority on the Jacobian power and the coercivity/domination constants; the routine integrability lemmas were spot-checked.

3 Fidelity / soundness findings

Sound; no substantive statement bug. The reviewer confirmed the Jacobian exponent $|R|^{-d} = |R^d|^{-1}$ with $d = \text{card } \iota$, becoming $(\sqrt{t})^d$ at $R = (\sqrt{t})^{-1}$, and its clean cancellation in the expectation/covariance ratios; the coercivity constants ($\frac{c}{2}\|u\|^2$ globally, $\frac{c}{\text{card } \iota} \sum_i u_i^2$ when passing

from the sup norm to the coordinate sum); and the elementary helpers. Two docstrings *over-claimed*: `integral_comp_smul_pi` advertised “for any nonzero R ” though the theorem carries no $R \neq 0$ hypothesis, and `abs_gaussianWeight_mul_exp_sub_one_le_uniform` claimed it makes the integrand “dominated by an *integrable* function” when it only states the pointwise bound (integrability of the gW summand needs positive-definiteness of H , supplied elsewhere). Both were corrected to match what the theorems actually deliver.

4 Simplifications applied

A dead open, six redundant opens, three dead equation-names. `open Module` was dead (its sole use `Module.finrank_pi` is qualified). More interestingly, the namespace-level `open MeasureTheory` persists through every `section` in the namespace, so the six per-section `open MeasureTheory` re-opens were *redundant* — each removable with the section still building. And three `set ...with hX` equation names (`hr_def`, two `hM_def`) were unused. All build green, signatures byte-identical.

What was *not* removed. Two non-vanishing locals (the $2\alpha \neq 0$ and $4\alpha \neq 0$ `haves`) sit immediately before a `field_simp` and supply exactly what it needs to clear denominators — `field_simp` fuel, consumed implicitly (the same trap as the `QuarticBoundedPrior` review’s positivity-consumed locals); kept. And several *exported* helpers that are unused *within this file* (`gaussianWeight_le_exp_neg_const_sq` and friends) are report-only: removing a public lemma changes the signature surface, which is out of a review tide’s scope (and they may be intended downstream API — a forward tide should check references first).

5 Disagreements and open questions

None left open — but the affirmation pass earned its keep. My first `integral_comp_smul_pi` docstring fix explained the no-hypothesis generality with “at $R = 0$ both sides are 0,” which the reviewer flagged as false in the $d = 0$ (empty-index) corner: the factor is $|0|^{-0} = 1$, not 0. I replaced it with the reviewer’s safer wording (“the $R = 0$ case is covered by the totalised-integral convention”). A docstring fix that itself slightly over-claimed, caught by the affirmation — exactly what the second pass is for.

6 Lessons

- **Namespace-level `open` makes per-section re-opens redundant.** A file that re-opens the same namespace at the top of every `section` is accreting harmless noise: the namespace-scoped `open` is already in force inside the sections. Removing the re-opens is a legitimate dead-open cleanup (build-confirmed), distinct from the usual “templated `open Real` never used.”
- **The affirmation pass is a real check on the reviewer’s *own* edits, not a rubber stamp.** A docstring “fix” can introduce a fresh inaccuracy (here a false $d = 0$ corner claim); sending the diff back for affirmation caught it. When a fix adds an *explanation*, the explanation needs the same scrutiny as the thing it explains.
- ***`_ne` locals before a `field_simp` are fuel, not dead.** Recurring on yet another file: a $\neq 0$ `have` that `grep` finds “unused by name” is consumed implicitly by the next `field_simp`. Read the few lines after it before deleting.

7 Follow-ups

- Forward tide: audit the exported-but-intra-file-unused helpers (`gaussianWeight_le_exp_neg_const_sq` *et al.*) for downstream references; prune or privatise the genuinely unused ones (a signature change, hence not a review edit). Also the pre-existing unused-hypothesis-in-type on `abs_rescaledObservable_linear_error_le`.
- The remaining never-reviewed laplace files are the `Multi/Covariance`, `CovarianceSharp`, and `CovarianceExplicit` trio (3998–19611 lines) — genuinely sectioned-review territory.